Crater Formation

Background

Impact craters are formed when pieces of asteroids or comets strike the surface of a planetary body. Craters are found on all the terrestrial planets, on the Earth's Moon, and on most satellites of planets. Various geological clues and studies of the lunar rocks returned by the Apollo missions indicate that about 3.9 billion years ago asteroid-size chunks of matter were abundant in the solar system. This was a time of intense bombardment of the young planets, affecting Earth by breaking up and modifying parts of the crust. Mountain building, plate tectonics, weathering and erosion have largely removed the traces of Earth's early cratering period. But the near absence of weathering on the Moon has allowed the evidence of this ancient time to be preserved.

Impact craters are formed by the transfer of energy from a moving mass (meteorite) to a stationary body (planet). Kinetic energy is the energy of motion. It is defined as one half the mass of an object, times the velocity of the object squared (K.E. = $\frac{1}{2}mv^2$).

Objects in space move very fast, so this can be a huge amount of energy! In an impact the kinetic energy of a meteorite is changed into heat that melts rocks and energy that pulverizes and excavates rock. Simplified demonstrations of this transfer of energy can be made by creating impacts in powdered materials. If identical objects are impacted into powdered materials from different heights or using different propulsion systems to increase velocities, then students can determine the effect velocity has on the cratering process. The high velocity impact and explosion of an iron meteorite about 30 meters in diameter could make a crater over one kilometre wide. This is how the Meteor Crater in Arizona was formed.

Aim: To investigate what happens to the size of craters as the velocity of the impact is increased.

Materials - **Per Group**

- marble
- ruler
- icecream container
- sand

Procedure

1. Collect the materials required.
2. Pour sand in the icecream container so that it is approximately one-third filled.
3. Drop the marble from 30 cm above the level of the sand.
4. Measure the depth and width of the crater formed in millimetres.
5. Record measurements in the table provided
6. Level the sand again and repeat steps 3 – 5 another two times.
7. Complete steps 3 – 6 for the other heights in the table.

Results

Height dropped from (cm)	Crater width (mm)				Crater Depth (mm)			
	Trial 1	Trial 2	Trial 3	Average	Trial 1	Trial 2	Trial 3	Average
30								
50								
80								
100								
150								

Discussion

1. What is the independent variable (the variable you are testing) in this experiment?

2. What is the dependent variable (the variable you are measuring) in this experiment?

3. What are some controlled variables (the ones that stayed the same) in the experiment?

4. Calculate the speed of the marble as it hits the sand for each height using the following formula:

   Speed = $\sqrt{0.196 \times height}$

Height (cm)	Speed (ms^{-1})
30	
50	
80	
100	
150	

5. Draw a graph of your results below

6. What happened to the size of the crater as the height the marble was dropped from was increased?

7. If you were going to conduct the experiment again, what could you do to improve it?